T. Miguel Pegues

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Education

Northwestern University, Evanston, IL — 2025 – December 2026 (expected)
M.S. in Robotics

California Institute of Technology, Pasadena, CA — 2018 – 2022
B.S. in Mechanical Engineering

Skills

- **Robotics:** Computer vision, manipulation, motion planning, ROS, ROS 2, MoveIt 2, SLAM
- **Software Development:** C++, Python, C, Git, Linux, unit testing, embedded programming
- **Mechanical:** CAD (Creo, SolidWorks), machining, design for assembly and manufacturing, 3D printing
- **Software Libraries:** CoppeliaSim, Gazebo, OpenCV, PyTorch, sklearn

Projects

Wearable Flexible Extra Finger (Polydact) — January - March 2026

- Created a tentacle-like extra finger to allows a user to manipulate a wider range of objects than they can naturally
- Designed and fabricated human interfacing parts for sensors, motors, control boards, and the digit itself
- Primary control device is a hand-made glove with integrated flex sensors interfacing with ROS 2 through a Pico 2 sending analog values to a Python ROS 2 node over USB or is read directly by an ADC connected to a Raspberry Pi 4
- Flex sensor values map to motor velocities, spooling and unspooling cables to flex the digit in all three directions

Dynamic Impedance Prosthetic Ankle — May 2026 - Present

- Designed ankle prosthesis with adjustable impedance to allow users to vary gait speed at will
- Built and tested novel mechanisms (radial cantilevers, magnetic return, among others) for adjusting stiffness of torsion-spring-like rotational elements under high-load cyclic use
- Implemented gait detection algorithms using IMU and force sensing to modulate quasi-static mechanism to automatically adjust prosthesis properties to the user's speed

Bug Catching Robot — December 2025

- Wrote ROS 2 packages for monocular object detection and motion planning to use a Franka Emika Robot arm to catch and sort colored HexBugs based on visual user input with a team of four
- Drove development of motion planning and execution for dynamic tracking of a moving target using MoveIt 2 and ros2_control, maximizing responsiveness by sending joint trajectories directly to the joint_trajectory_controller
- Designed and 3D printed parts for physical arena and supported camera calibration with Python, OpenCV, and ROS 2

Jenga Playing Robot — Winter - Spring 2022

- Designed and built a robot arm capable of playing Jenga with a team of four
- Fabricated metal robot parts, Jenga block markers, and alignment tools after designing in SolidWorks
- Created and tuned OpenCV vision system to locate 54 individual Jenga blocks in 3D space with millimeter precision and send coordinates in ROS message to motion planning node

Work Experience

Mechanical Engineering Intern — Summer 2026
Productive Robotics – Carpinteria, CA

- Prototyped components and accessories for Productive's existing 7-axis arms and unreleased industrial robotic products
- Designed, built, and wrote software for robotic fixtures to automate internal manufacturing quality checks, increasing production uptime and reducing human error

Mechanical Structures Engineer — July 2022 - July 2025
Whirlpool Corporation – Benton Harbor, MI

- Drove Cost & Quality and Product Development projects, designing plastic and metal components for all refrigeration architectures to improve cost and customer satisfaction
- Managed verification and validation testing for individual components, systems, and products